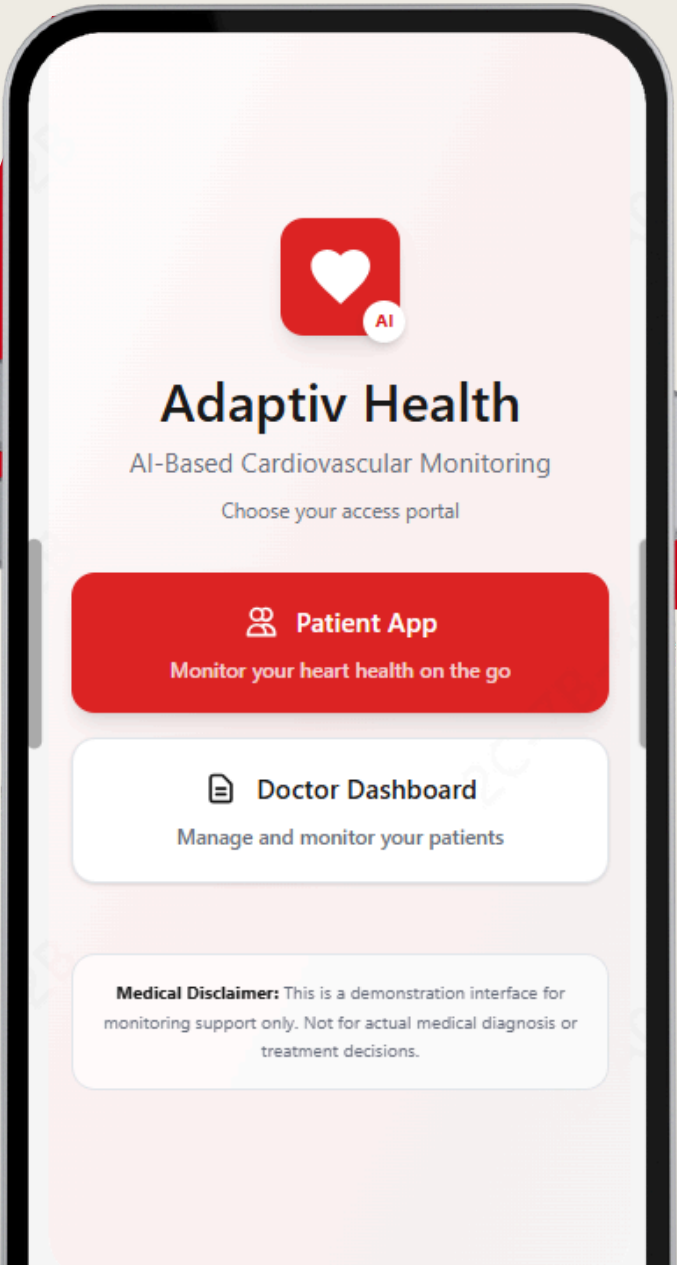


AI-POWERED CARDIOVASCULAR MONITORING SYSTEM

ADAPTIV HEALTH



ADAPTIV HEALTH —
SMARTER MOVEMENT,
SAFER HEARTS.

PRESENTED BY
PULSE.

PRESENTED BY
TEAM PULSE



TEAM PULSE

— BRINGS TOGETHER DIFFERENT SKILLS, ONE MISSION, AND A SHARED HEARTBEAT FOR INNOVATION.

ABDALLA HEGGY #8457785

OMAR MASSOUD #7858681

GHINA ISHAQ #871179

KARYLLE OCHAVE #7669057

BITHIN RADHAKRISHNAN #8910881

HAMZA AL RAJABY #7322586

(02)

Team Pulse is a diverse group of innovators combining AI, cybersecurity, and business insight to build heart-safe digital health solutions.



THE PROBLEM

**17.9 MILLION PEOPLE DIE FROM CARDIOVASCULAR DISEASE EVERY YEAR — THE WORLD'S #1 KILLER
YET THE CRITICAL GAP BETWEEN HOSPITAL DISCHARGE AND DAILY LIFE REMAINS UNSOLVED.**

No Real-Time Monitoring

Patients leave hospital with zero continuous oversight between appointments.

Blind Spots for Doctors

Clinicians cannot see what happens to patients outside the clinic walls.

Fitness Apps Fall Short

Generic apps ignore cardiac safety entirely — one wrong workout can be fatal.

**ADAPTIV HEALTH —
SMARTER MOVEMENT,
SAFER HEARTS.**

OUR SOLUTION



BRIDGING THE GAP BETWEEN CLINIC AND DAILY LIFE

NOT A FITNESS APP. NOT A DIAGNOSTIC TOOL. A SAFETY NET FOR CARDIAC PATIENTS.

COLLECT VITALS



REAL-TIME RISK CLASSIFICATION

CONTINUOUS CARDIOVASCULAR RISK SCORING, COLOUR-CODED FOR INSTANT CLARITY.

GENERATE AI SCORE



PERSONALISED AI EXERCISE GUIDANCE

SAFE, TAILORED WORKOUT PLANS GENERATED FROM LIVE CARDIAC DATA.

DELIVER EXERCISE PLAN



INSTANT DANGER ALERTS

AUTOMATIC ALERTS TO CLINICIANS THE MOMENT READINGS BECOME DANGEROUS.

TRIGGER DOCTOR ALERT



WORKS COMPLETELY OFFLINE

ON-DEVICE EDGE AI ENSURES PROTECTION EVEN WITHOUT INTERNET CONNECTIVITY.

**ADAPTIV HEALTH —
SMARTER MOVEMENT,
SAFER HEARTS.**



SYSTEM ARCHITECTURE

FOUR LAYERS. ONE SEAMLESS SYSTEM.
FULLY DEPLOYED AT [API.BACK-ADAPTIVEHEALTHUOWD.XYZ](https://api.back-adaptivehealthuowd.xyz) — AWS EC2 MUMBAI, PRIVATE VPC, POSTGRESQL RDS.

STACK OVERVIEW

01

FLUTTER APP (EDGE AI)

PATIENT-FACING MOBILE APP WITH ON-DEVICE INFERENCE.

02

AWS API GATEWAY + FASTAPI

ASYNCR PRODUCTION-GRADE BACKEND ON EC2, MUMBAI REGION.

03

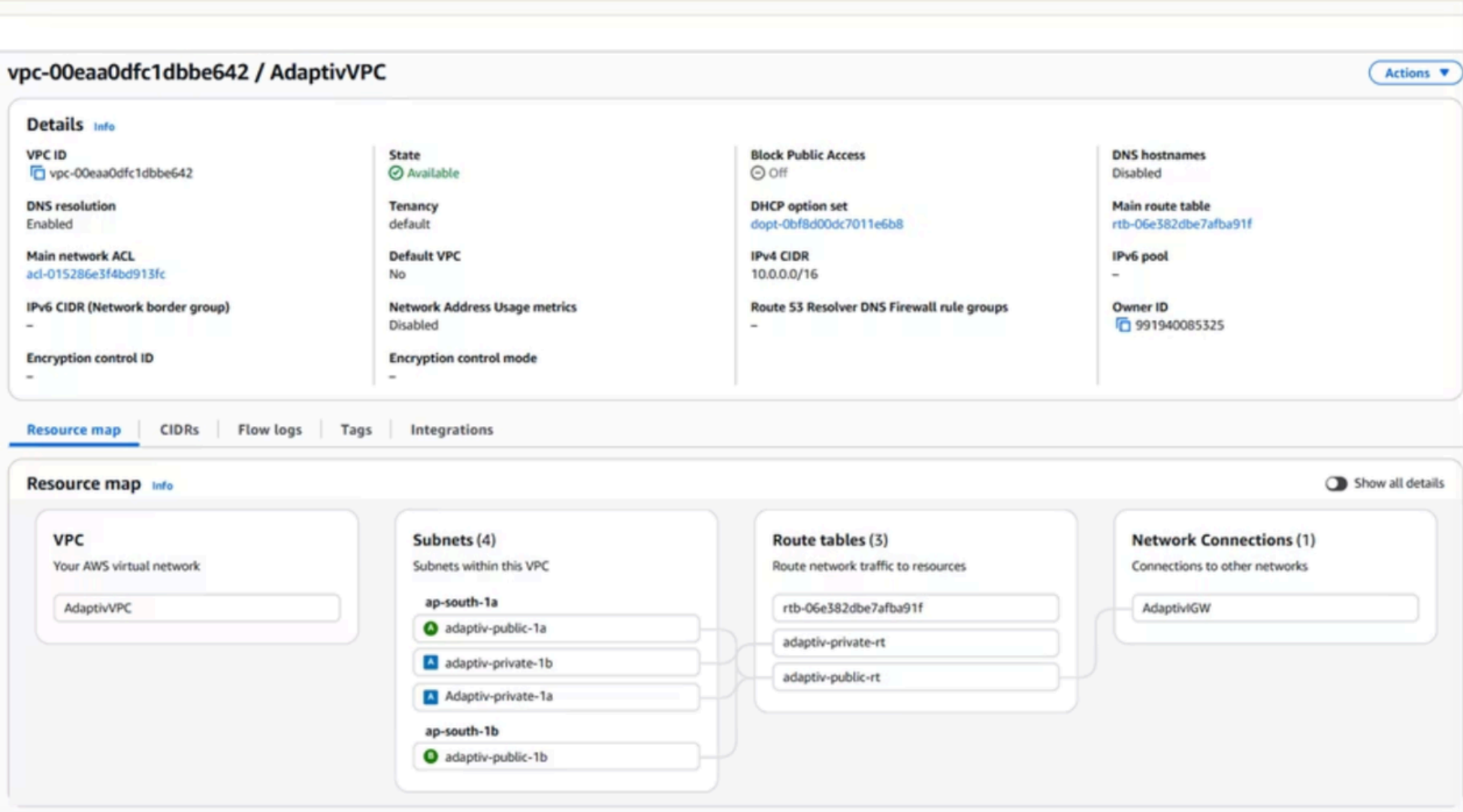
POSTGRESQL RDS (PRIVATE VPC)

ZERO PUBLIC INTERNET ACCESS — ALL PHI ENCRYPTED AT REST.

04

REACT CLINICIAN DASHBOARD

REAL-TIME OVERSIGHT FOR DOCTORS FROM ANY BROWSER.



PATIENT APP



THE PATIENT'S SAFETY COMPANION

→ LIVE VITALS DISPLAY

REAL-TIME HR, SPO2, AND BLOOD PRESSURE WITH COLOUR-CODED SAFETY ZONES (GREEN / AMBER / RED).

→ ON-DEVICE EDGE AI

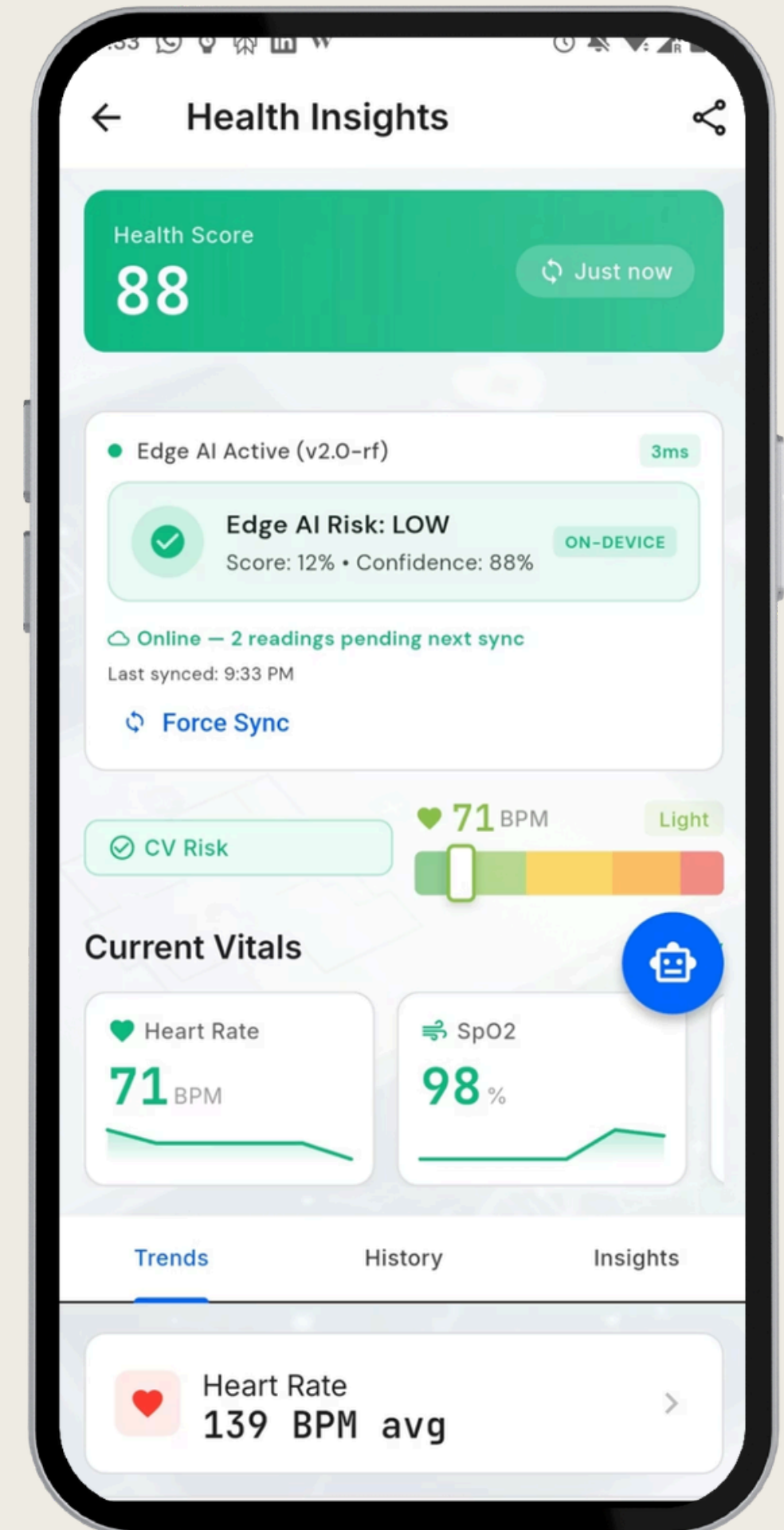
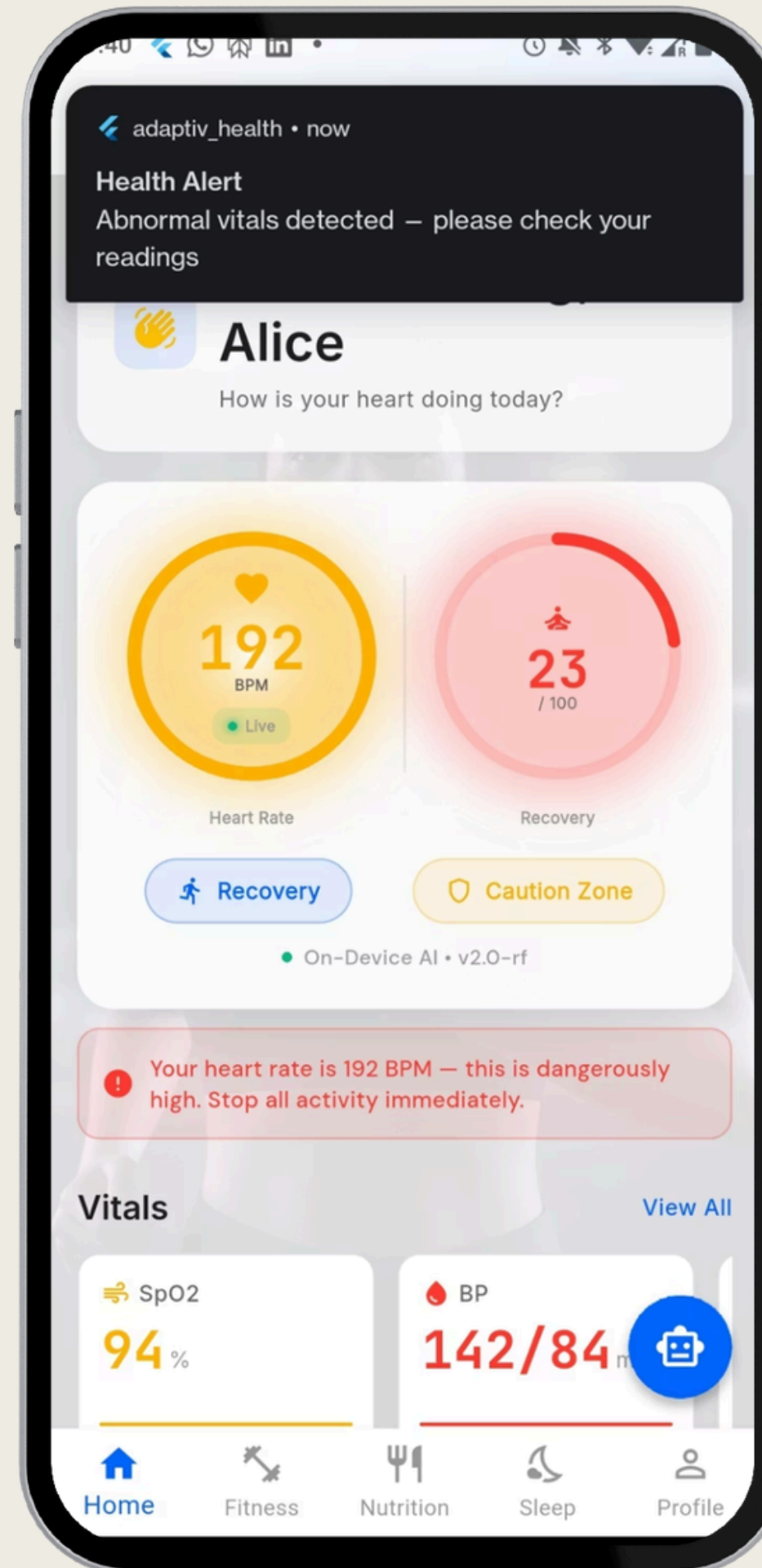
PURE-DART RANDOM FOREST MODEL — WORKS COMPLETELY OFFLINE, <160MS INFERENCE.

→ BLE WEARABLE + GEMINI COACH

BLUETOOTH WEARABLE CONNECTIVITY AND AI HEALTH COACH POWERED BY GEMINI FOR NARRATIVE GUIDANCE.

→ OFFLINE SYNC + SECURITY

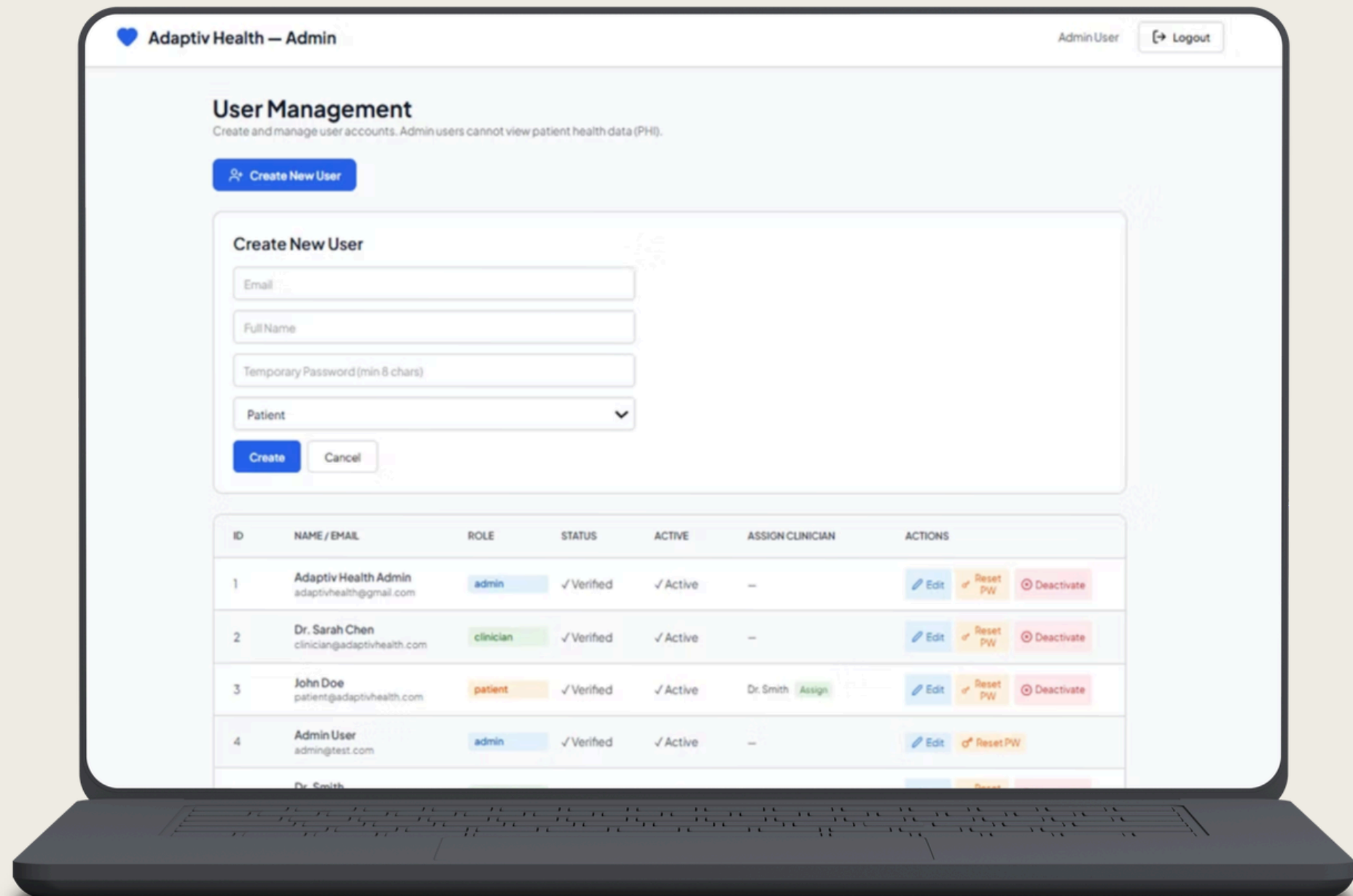
AUTOMATIC SYNC ON RECONNECTION; BRUTE FORCE PROTECTION LOCKS ACCOUNT AFTER FAILED ATTEMPTS.



CLINICIAN DASHBOARD



CLINICAL OVERSIGHT, ANYWHERE, ANYTIME



LIVE PATIENT LIST

REAL-TIME HEALTH SCORES
AND RISK LEVELS ACROSS
ALL ASSIGNED PATIENTS.

AI RISK SUMMARIES

AI-GENERATED NARRATIVE
SUMMARIES AND
INTERACTIVE
CARDIOVASCULAR TREND
GRAPHS.

CRITICAL ALERT PANEL

COLOUR-CODED SEVERITY
ALERTS AND POPULATION-
LEVEL ANALYTICS AT A
GLANCE.

ROLE-BASED ACCESS

SECURE MESSAGING AND
STRICTLY ENFORCED
DOCTOR / ADMIN / PATIENT
ACCESS TIERS.

AI & MACHINE LEARNING

THREE LAYERS OF INTELLIGENCE

LAYER 1 — EDGE AI (ON-DEVICE)

RANDOM FOREST · PURE DART ·
<160MS · FULLY OFFLINE

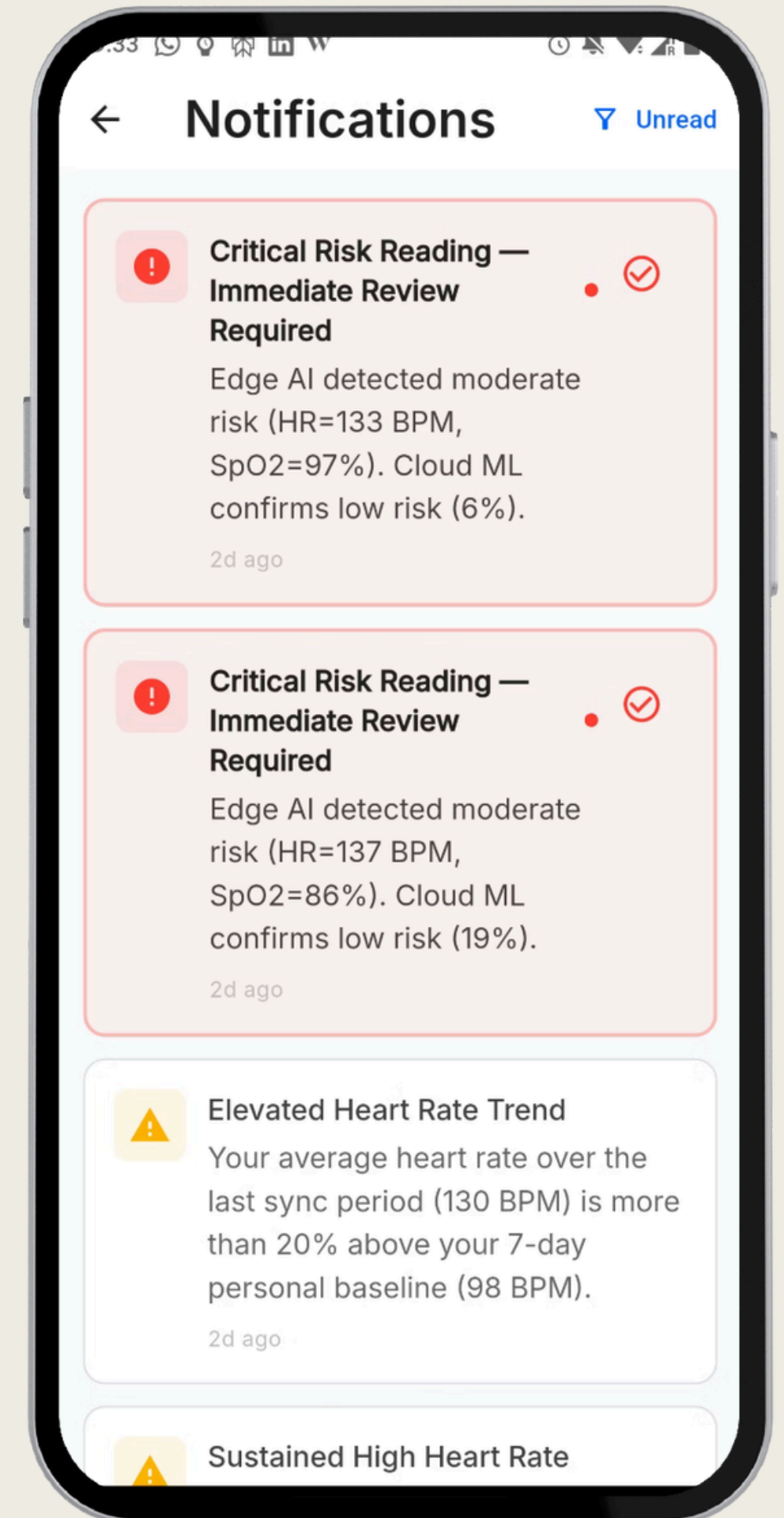
LAYER 2 — CLOUD ML (AWS EC2)

RANDOM FOREST · SCIKIT-
LEARN · 96.9% ACCURACY · HIGH
RISK RECALL OPTIMISED

LAYER 3 — GEMINI AI COACH

NARRATIVE RECOMMENDATIONS
· TEMPERATURE 0.4 · DUAL-PATH
FALLBACK — NEVER OFFLINE

RISK CLASSIFICATION: LOW (0.0–0.49) → CONTINUE SAFELY · MODERATE (0.50–0.79) → SLOW DOWN · HIGH (0.80–1.0) → STOP IMMEDIATELY + ALERT DOCTOR





SECURITY

HEALTHCARE-GRADE SECURITY. NO COMPROMISES.



ARGON2ID PASSWORD HASHING

OWASP & NIST SP 800-63B RECOMMENDED STANDARD.



AES-256-GCM ENCRYPTION

EVERY PHI FIELD ENCRYPTED BEFORE DATABASE STORAGE.



JWT WITH JTI BLOCKLIST

IMMEDIATE TOKEN REVOCATION ON LOGOUT — STATEFUL SESSION CONTROL.



AWS PRIVATE VPC

DATABASE HAS ZERO PUBLIC INTERNET ACCESS.



HIPAA & UAE DHA COMPLIANT

CLOUDTRAIL AUDIT LOGS WITH 6-YEAR RETENTION.



ROLE-BASED ACCESS CONTROL

PATIENT · DOCTOR · ADMIN — STRICTLY ENFORCED ACROSS ALL ENDPOINTS.



TEST RESULTS

46 TEST CASES. 100% PASS RATE.

TESTED AGAINST THE LIVE PRODUCTION SYSTEM AT [API.BACK-ADAPTIVHEALTHUOWD.XYZ](https://api.back-adaptivhealthuowd.xyz)

Category	Tests	Result
Functionality	12	PASS
Database	8	PASS
UX / UI	10	PASS
Security	9	PASS
Machine Learning	7	PASS
TOTAL	46	100% ✓

46

TOTAL TESTS
ACROSS 5 CATEGORIES

100%

PASS RATE
ZERO FAILURES ON LIVE SYSTEM

EVOLUTION



WHAT WE BUILT VS. WHAT WE PLANNED

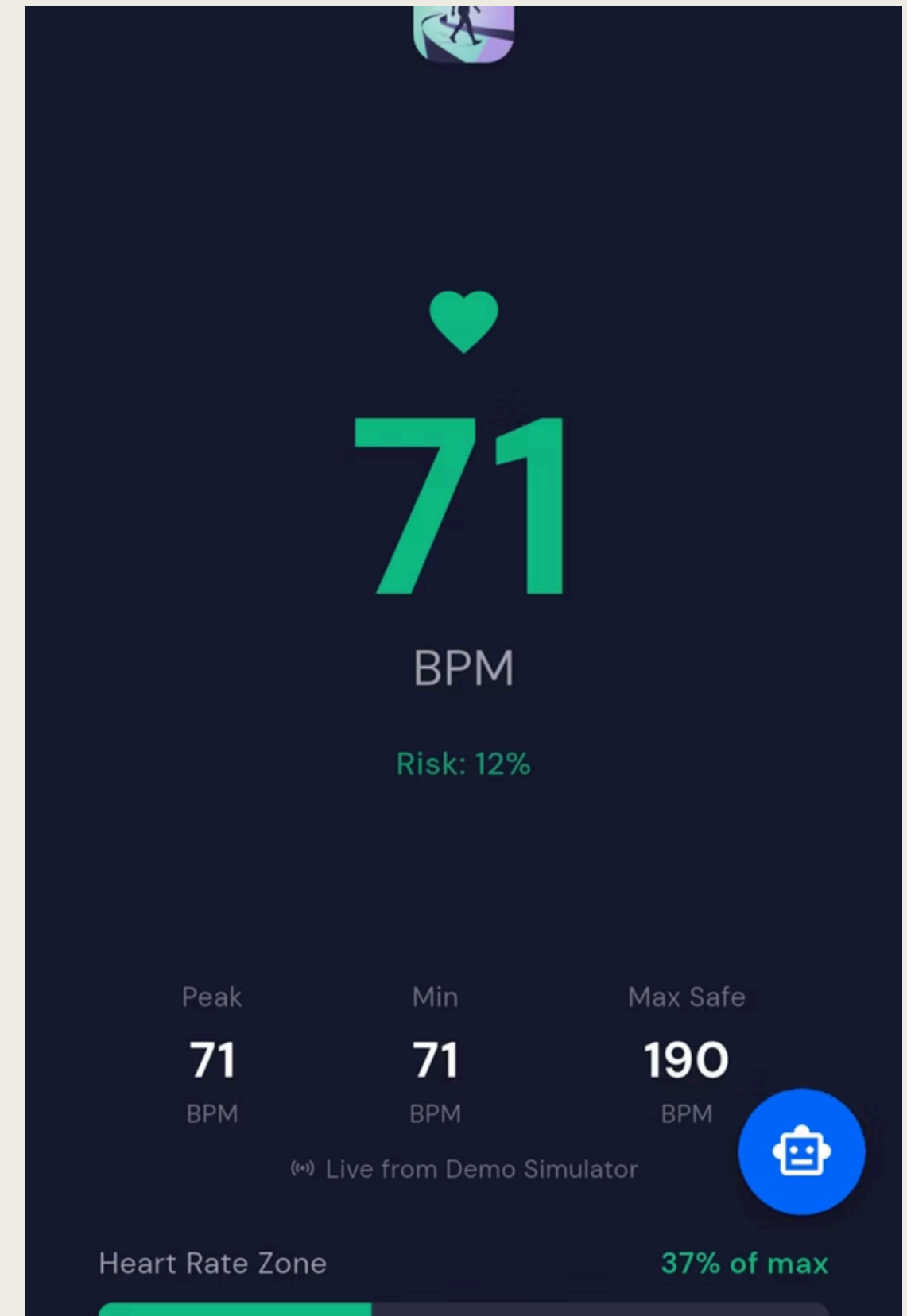
WE DIDN'T JUST PLAN IT — WE CHANGED AND IMPROVED IT AT EVERY STAGE.

PLANNED → DELIVERED

FLASK → FASTAPI (ASYNC, PRODUCTION-GRADE) TENSORFLOW/LSTM → RANDOM FOREST (MEDICALLY VALIDATED) FIREBASE + AWS → AWS RDS ONLY (HIPAA COMPLIANT) BCrypt → ARGON2ID (OWASP GOLD STANDARD) BASIC JWT → JWT + JTI BLOCKLIST TFLITE → PURE-DART (TRULY OFFLINE, ZERO DEPENDENCIES)

ADDED IN SEMESTER 2

◆ GEMINI AI HEALTH COACH ◆ ADMIN DASHBOARD ◆ OFFLINE SYNC QUEUE ◆ CLOUDTRAIL HIPAA AUDIT LOGGING

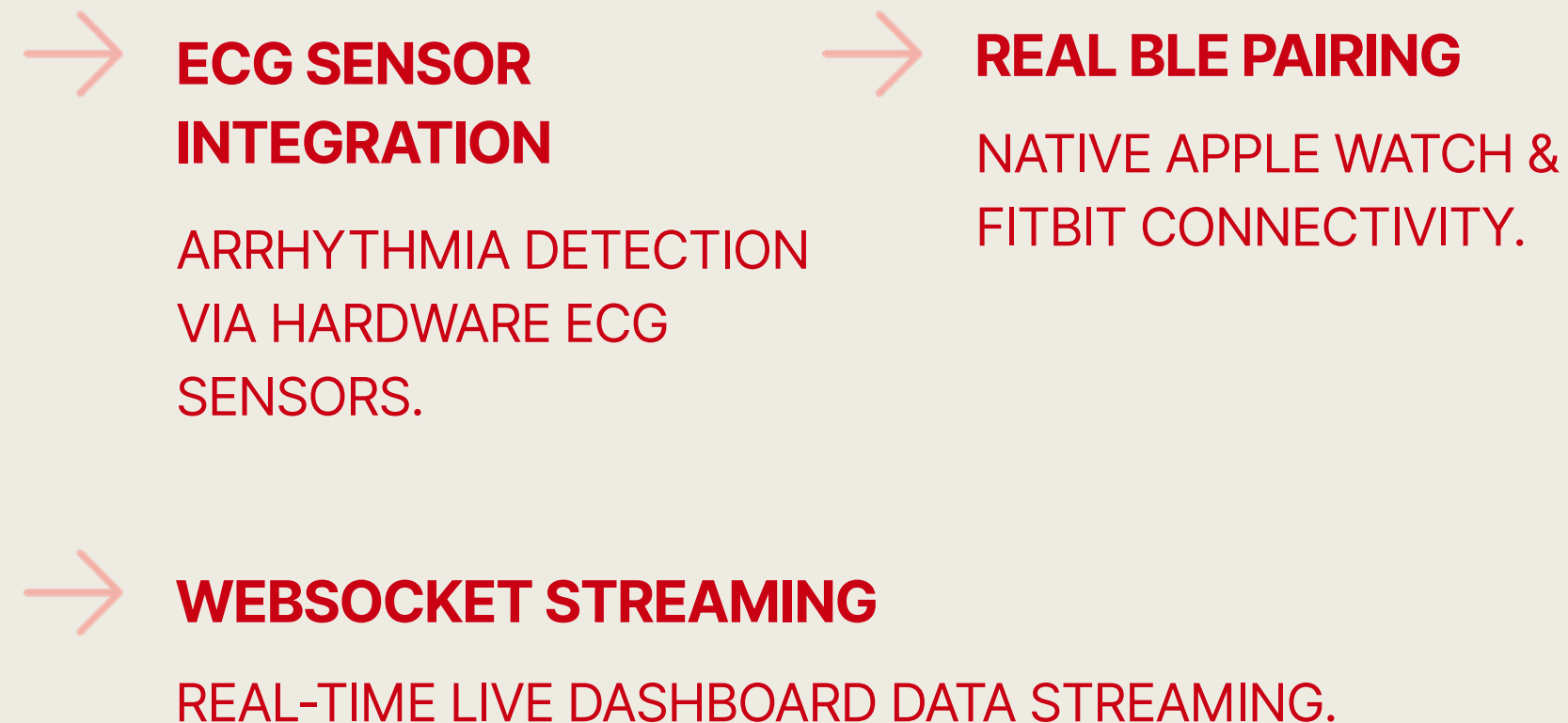




FUTURE WORK

THE FOUNDATION IS BUILT. HERE'S WHAT COMES NEXT.

NEAR-TERM ROADMAP



LONG-TERM VISION





KEEPING CARDIAC PATIENTS SAFE — EVERY HEARTBEAT.

LIVE SYSTEM

DASHBOARD-
ADAPTIVHEALTHUOWD.XYZ

TEAM

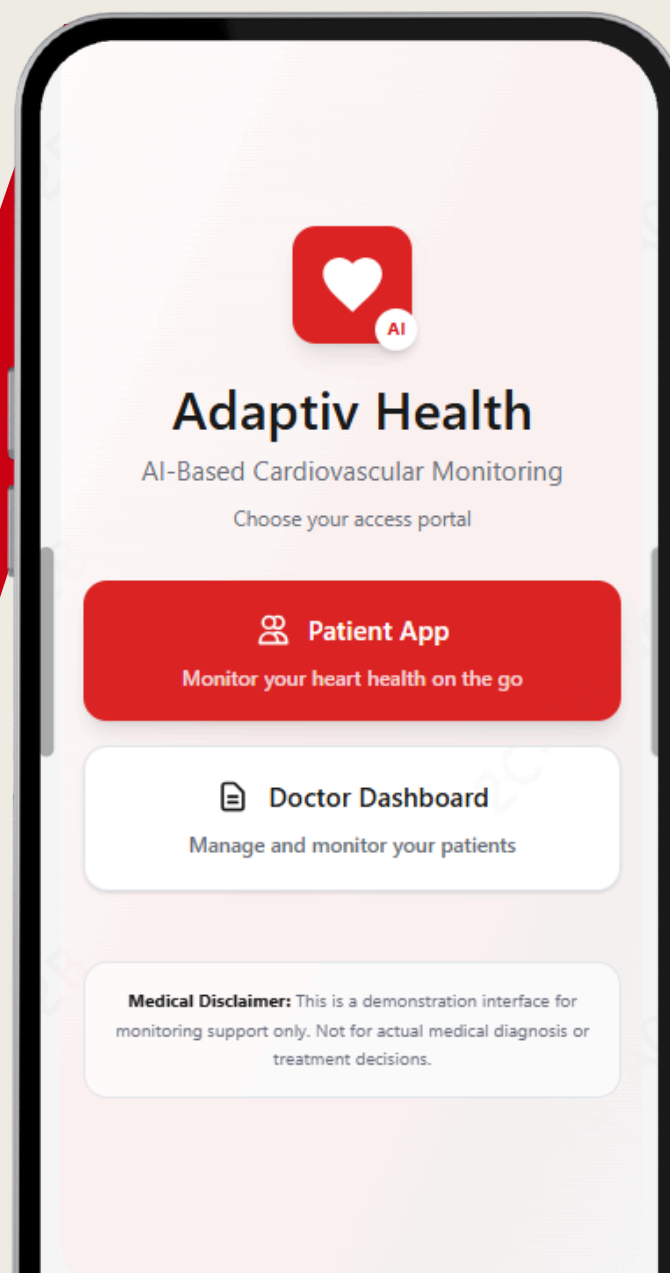
TEAM PULSE · CSIT321 ·
UOWD 2026

QUESTIONS

WE'D LOVE TO WALK YOU THROUGH THE LIVE SYSTEM.



ADAPTIV HEALTH



ADAPTIV HEALTH —
SMARTER MOVEMENT,
SAFER HEARTS.

PRESENTED BY
PULSE.